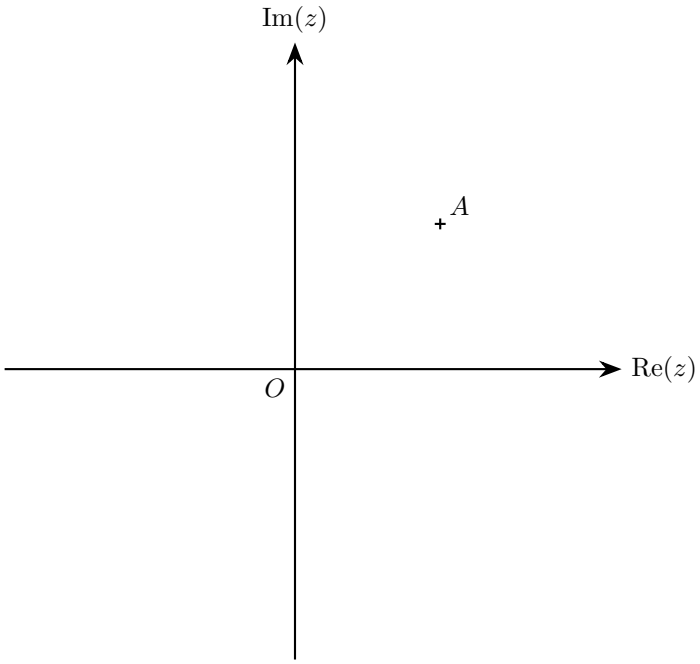
CORE PURE / COMPLEX NUMBERS

# Complex Roots and Geometry Problems

Worked solutions 13 questions · 126 marks

## Questions


| | | |
|---|---|---|
| Question 1 | *8 marks* | 2 |
| Question 2 | *10 marks* | 4 |
| Question 3 | *9 marks* | 6 |
| Question 4 | *10 marks* | 8 |
| Question 5 | *5 marks* | 10 |
| Question 6 | *10 marks* | 12 |
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| Question 11 | *14 marks* | 25 |
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**1.** An Argand diagram with the point $A$ representing a complex number $z_1$ is shown in the diagram. The complex numbers $z_2$, $z_3$ and $z_4$ are $z_1 e^{\frac{1}{2}\pi i}$, $z_1 e^{\pi i}$ and $z_1 e^{\frac{3}{2}\pi i}$ respectively.

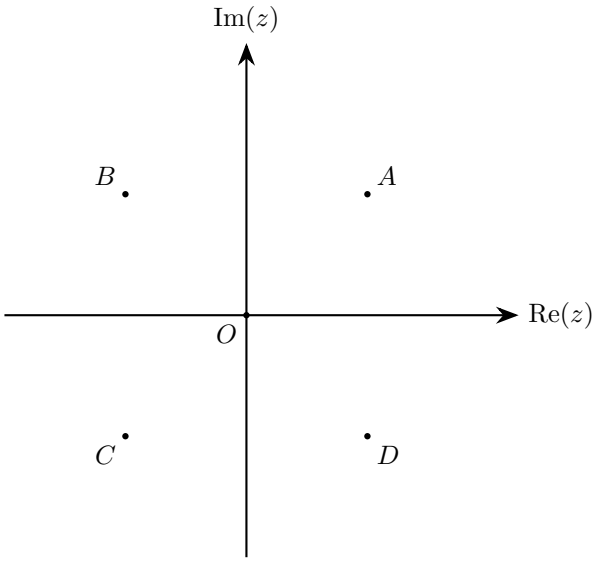
**(a)** (i) On a copy of the Argand diagram, mark the points $B$, $C$ and $D$ representing the complex numbers $z_2$, $z_3$ and $z_4$. **[2]**

(ii) Show that $z_1 + z_2 + z_3 + z_4 = 0$. **[2]**

**(b)** Given now that $z_1$, $z_2$, $z_3$ and $z_4$ are roots of the equation $z^4 = -16$, find these four roots, giving your answers in the form $a + ib$, where $a$ and $b$ are real and exact. **[4]**

**Solution**

**(a)** (i) Multiplying $z_1$ by $e^{i\theta}$ corresponds to rotating $z_1$ by $\theta$ anticlockwise about the origin.
So $B$, $C$ and $D$ are the images of $A$ under such rotations of $90°, 180°$ and $270°$ respectively.



(ii)

$$\begin{aligned} z_1 + z_2 + z_3 + z_4 &= z_1 + z_1 e^{\frac{1}{2}\pi i} + z_1 e^{\pi i} + z_1 e^{\frac{3}{2}\pi i} \\ &= z_1 + iz_1 - z_1 - iz_1 \\ &= z_1(1 + i - 1 - i) \\ &= 0 \end{aligned}$$

Hence $z_1 + z_2 + z_3 + z_4 = 0$.

(Alternatively, use that $1 + e^{\frac{1}{2}\pi i} + e^{\pi i} + e^{\frac{3}{2}\pi i}$ is a geometric sum with common ratio $r = e^{\frac{1}{2}\pi i}$)

**(b)** Write

$$\begin{aligned} -16 &= 16e^{i\pi} \\ &= 16e^{i(\pi + 2k\pi)} \end{aligned}$$

so the fourth roots are

$$z = 2e^{i\left(\frac{\pi + 2k\pi}{4}\right)} \qquad k = 0, 1, 2, 3$$

Hence

$$\begin{aligned} k = 0: \quad z &= 2e^{i\pi/4} = 2\left(\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right) = \sqrt{2} + i\sqrt{2} \\ k = 1: \quad z &= 2e^{3i\pi/4} = 2\left(-\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right) = -\sqrt{2} + i\sqrt{2} \\ k = 2: \quad z &= 2e^{5i\pi/4} = 2\left(-\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}\right) = -\sqrt{2} - i\sqrt{2} \\ k = 3: \quad z &= 2e^{7i\pi/4} = 2\left(\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}\right) = \sqrt{2} - i\sqrt{2} \end{aligned}$$

Therefore the four roots are

$$\sqrt{2} + i\sqrt{2}, \quad -\sqrt{2} + i\sqrt{2}, \quad -\sqrt{2} - i\sqrt{2}, \quad \sqrt{2} - i\sqrt{2}$$

**2.** The five solutions of the equation $z^5 = 16 + 16\sqrt{3}\,i$ are represented by points on an Argand diagram.

These points are the vertices of a regular pentagon $P$ with centre at the origin.

(a) Determine the five complex numbers that represent these vertices, giving your answers in the form $re^{i\theta}$, where $0 \leq \theta < 2\pi$. **[4]**

(b) Sketch $P$ on an Argand Diagram, taking care to label each vertex. **[2]**

(c) Show that the area of $P$ can be written $k \sin \frac{2\pi}{5}$ where $k$ is an integer to be determined. **[4]**

**Solution**

**(a)** Let

$$w = 16 + 16\sqrt{3}\,i$$

First write $w$ in polar form.

Its modulus is

$$\begin{aligned} |w| &= \sqrt{16^2 + (16\sqrt{3})^2} \\ &= \sqrt{256 + 768} \\ &= \sqrt{1024} \\ &= 32 \end{aligned}$$

Its argument is

$$\arg(w) = \tan^{-1}\left(\frac{16\sqrt{3}}{16}\right) = \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$$

since $w$ lies in the first quadrant.

So

$$16 + 16\sqrt{3}\,i = 32e^{i\pi/3}$$

Now solve

$$z^5 = 32e^{i\pi/3}$$

The fifth roots have modulus

$$32^{1/5} = 2$$

and arguments

$$\theta = \frac{\frac{\pi}{3} + 2\pi k}{5} = \frac{\pi}{15} + \frac{2\pi k}{5} \qquad (k = 0, 1, 2, 3, 4)$$

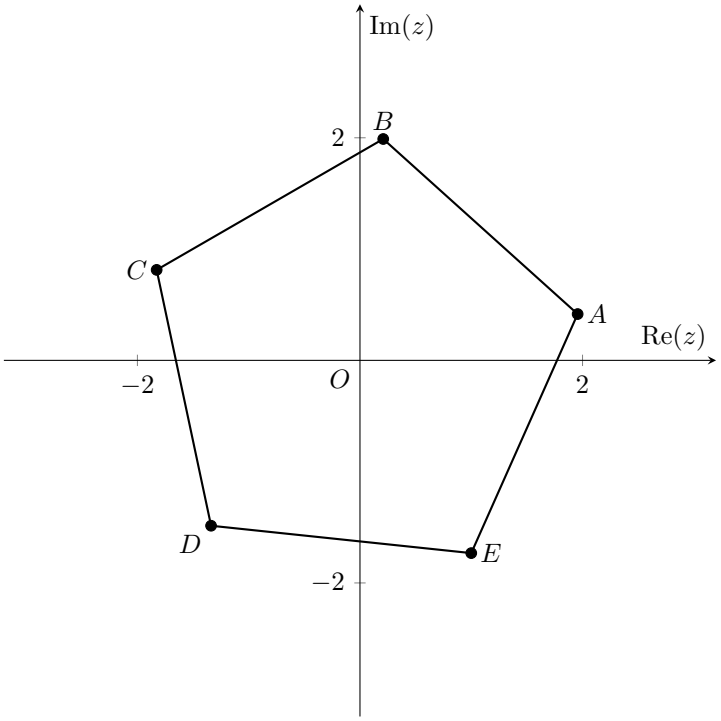
Hence the five roots are

$$\begin{aligned} z &= 2e^{i\pi/15} \\ z &= 2e^{i7\pi/15} \\ z &= 2e^{i13\pi/15} \\ z &= 2e^{i19\pi/15} \\ z &= 2e^{i25\pi/15} = 2e^{i5\pi/3} \end{aligned}$$

**(b)** Let the vertices in anticlockwise order be

$$A = 2e^{i\pi/15}, \quad B = 2e^{i7\pi/15}, \quad C = 2e^{i13\pi/15}, \quad D = 2e^{i19\pi/15}, \quad E = 2e^{i5\pi/3}$$

They all lie on the circle of radius 2 centred at the origin, with equal angular spacing $\frac{2\pi}{5}$.




**(c)** Join the origin to each vertex. This splits the pentagon into 5 congruent isosceles triangles.
Each vertex has modulus 2, so each triangle has two sides of length 2.
Also, the arguments of adjacent vertices differ by

$$\frac{2\pi}{5}$$

so the angle at the origin in each triangle is $\frac{2\pi}{5}$.
Hence the area of one triangle is

$$\frac{1}{2} \times 2 \times 2 \times \sin\frac{2\pi}{5} = 2\sin\frac{2\pi}{5}$$

Therefore the area of the pentagon $P$ is

$$5\left(2\sin\frac{2\pi}{5}\right) = 10\sin\frac{2\pi}{5}$$

So it can be written in the form $k\sin\frac{2\pi}{5}$ with

$$k = 10$$

**3.** In an Argand diagram, the points $A$ and $B$ are represented by the complex numbers $1+4i$ and $7+2i$ respectively. The circle $C$ passes through $A$ and $B$, and its centre lies on the real axis.

**(a)** Find the equation of $C$, giving your answer in the form

$$|z-a|=b \qquad a\in\mathbb{C},\ b\in\mathbb{R}$$

**[4]**

The circle $D$, with equation $|z-(1+i)|=5$, intersects $C$ at the points representing the complex numbers $z_1$ and $z_2$.

**(b)** Find the complex numbers $z_1$ and $z_2$. **[5]**

**Solution**

**(a)** Let the centre of circle $C$ be the point represented by the real number $h$, since the centre lies on the real axis.

So the centre is $(h,0)$ on the Argand diagram.

Since $A(1,4)$ and $B(7,2)$ both lie on $C$, their distances from the centre are equal:

$$(h-1)^2+4^2=(h-7)^2+2^2$$
$$(h-1)^2+16=(h-7)^2+4$$

Expanding:

$$h^2-2h+1+16=h^2-14h+49+4$$
$$-2h+17=-14h+53$$
$$12h=36$$
$$h=3$$

So the centre is the complex number 3.

Now find the radius using point $A$:

$$\begin{aligned} r^2&=(3-1)^2+4^2\\ &=2^2+4^2\\ &=4+16\\ &=20\end{aligned}$$

Hence

$$r=\sqrt{20}=2\sqrt{5}$$

Therefore the equation of $C$ is

$$|z-3|=2\sqrt{5}$$

**(b)** Let

$$z=x+iy$$

Since $z$ lies on $C$, we have

$$|z-3|=2\sqrt{5}$$

so

$$(x-3)^2+y^2=20$$

Since $z$ also lies on $D$, whose equation is $|z-(1+i)|=5$, we have

$$(x-1)^2+(y-1)^2=25$$

So the points of intersection satisfy the simultaneous equations

$$(x-3)^2 + y^2 = 20$$
$$(x-1)^2 + (y-1)^2 = 25$$

Expand the second equation:

$$x^2 - 2x + 1 + y^2 - 2y + 1 = 25$$
$$x^2 + y^2 - 2x - 2y - 23 = 0$$

Expand the first equation:

$$x^2 - 6x + 9 + y^2 = 20$$
$$x^2 + y^2 - 6x - 11 = 0$$

Subtract the first from the second:

$$(x^2 + y^2 - 2x - 2y - 23) - (x^2 + y^2 - 6x - 11) = 0$$
$$-2x - 2y - 23 + 6x + 11 = 0$$
$$4x - 2y - 12 = 0$$
$$2x - y - 6 = 0$$

Hence

$$y = 2x - 6$$

Substitute into $(x-3)^2 + y^2 = 20$:

$$(x-3)^2 + (2x-6)^2 = 20$$
$$(x-3)^2 + 4(x-3)^2 = 20$$
$$5(x-3)^2 = 20$$
$$(x-3)^2 = 4$$

So

$$x - 3 = \pm 2$$

giving

$$x = 1 \quad \text{or} \quad x = 5$$

Now use $y = 2x - 6$.
If $x = 1$,

$$y = 2(1) - 6 = -4$$

If $x = 5$,

$$y = 2(5) - 6 = 4$$

Therefore the two intersection points are represented by

$$z_1 = 1 - 4i, \qquad z_2 = 5 + 4i$$

**4.** Given that $z_1 = 7 - i$ and $z_2 = \dfrac{z_1}{1-i}$,

**(a)** find $z_2$ in the form $a + bi$, where $a$ and $b$ are real. **[2]**

**(b)** Show on an Argand diagram the point $P$ representing $z_1$ and the point $Q$ representing $z_2$. **[2]**

**(c)** Given that $O$ is the origin, show that $\angle OQP = \dfrac{\pi}{2}$. **[2]**

The circle passing through the points $O$, $P$ and $Q$ has centre $C$. Find

**(d)** the complex number represented by $C$. **[2]**

**(e)** the exact value of the radius of the circle. **[2]**

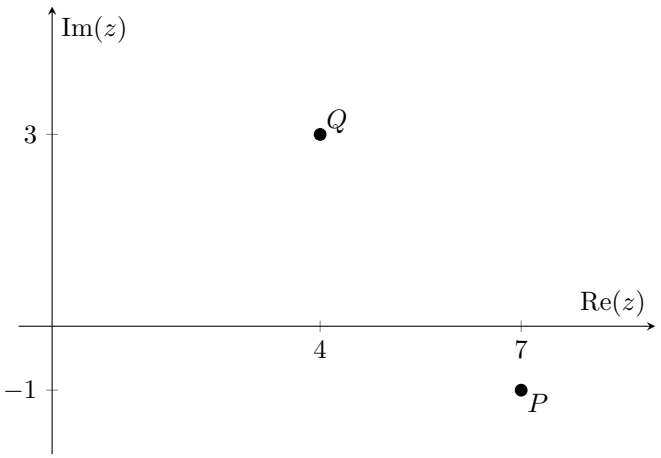
**Solution**

**(a)** Rationalise the denominator:

$$\begin{aligned} z_2 &= \frac{7-i}{1-i} \cdot \frac{1+i}{1+i} \\ &= \frac{(7-i)(1+i)}{(1-i)(1+i)} \\ &= \frac{7+7i-i-i^2}{1-i^2} \\ &= \frac{7+6i+1}{1+1} \\ &= \frac{8+6i}{2} \\ &= 4+3i \end{aligned}$$

**(b)** On an Argand diagram, the complex number $x + yi$ is represented by the point $(x, y)$. So

$$z_1 = 7 - i \;\Rightarrow\; P(7,-1), \qquad z_2 = 4 + 3i \;\Rightarrow\; Q(4,3)$$




**(c)** Using the coordinates from part (b),

$$O = (0,0), \quad P = (7,-1), \quad Q = (4,3)$$

Take the vectors from $Q$:

$$\overrightarrow{QO} = \begin{pmatrix} -4 \\ -3 \end{pmatrix}, \qquad \overrightarrow{QP} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$$

Their scalar product is

$$\begin{aligned}\overrightarrow{QO} \cdot \overrightarrow{QP} &= (-4)(3) + (-3)(-4) \\ &= -12 + 12 \\ &= 0\end{aligned}$$

So $\overrightarrow{QO}$ is perpendicular to $\overrightarrow{QP}$, which means

$$\angle OQP = \frac{\pi}{2}$$

**(d)** Since $\angle OQP = \dfrac{\pi}{2}$, the side $OP$ is a diameter of the circle through $O$, $P$ and $Q$.

Therefore the centre $C$ is the midpoint of $O(0,0)$ and $P(7,-1)$:

$$\begin{aligned}C &= \left(\frac{0+7}{2}, \frac{0+(-1)}{2}\right) \\ &= \left(\frac{7}{2}, -\frac{1}{2}\right)\end{aligned}$$

So the complex number represented by $C$ is

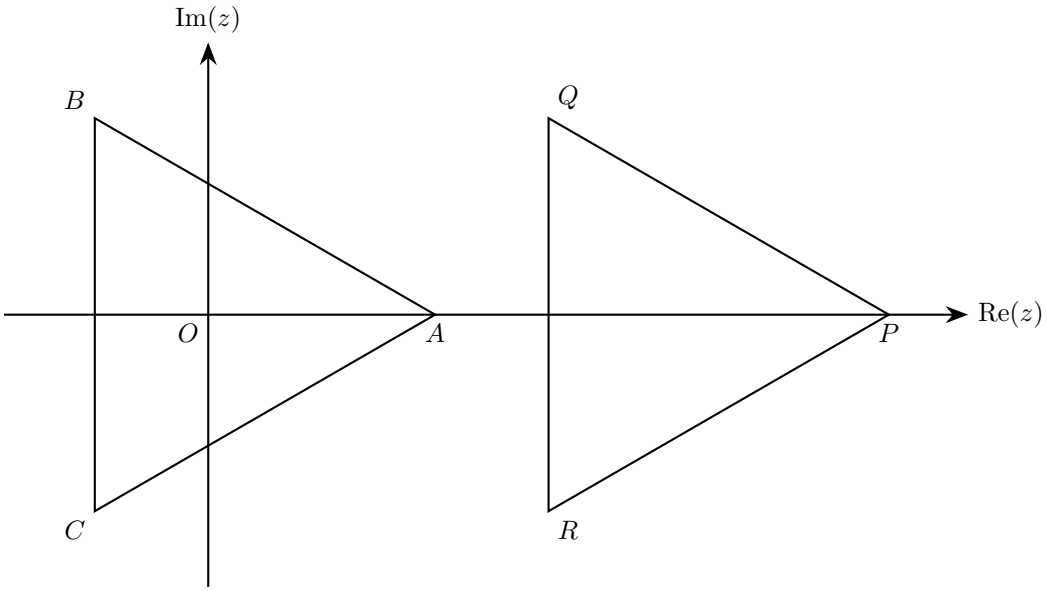
$$\frac{7}{2} - \frac{1}{2}i$$

**(e)** The diameter is $OP$, so first find its length:

$$\begin{aligned}OP &= \sqrt{(7-0)^2 + (-1-0)^2} \\ &= \sqrt{49+1} \\ &= \sqrt{50} \\ &= 5\sqrt{2}\end{aligned}$$

Hence the radius of the circle is

$$\frac{1}{2}OP = \frac{5\sqrt{2}}{2}$$




**5.** The cube roots of unity are represented on the Argand diagram above by the points $A$, $B$ and $C$.

The complex numbers represented by $P$, $Q$ and $R$ are obtained by adding 2 to the complex numbers represented by $A$, $B$ and $C$ respectively.

Determine a degree 6 polynomial equation with integer coefficients whose roots are the complex numbers represented by the points $A$, $B$, $C$, $P$, $Q$ and $R$. **[5]**

**Solution**

Let the required polynomial have variable $z$.
The points $A, B, C$ represent the cube roots of unity, so their complex numbers satisfy

$$z^3 = 1$$

Hence $A, B, C$ are the roots of

$$z^3 - 1 = 0$$

Now $P, Q, R$ are obtained by adding 2 to the complex numbers at $A, B, C$.
So if $z$ represents one of $P, Q, R$, then $z - 2$ is a cube root of unity. Therefore

$$(z-2)^3 = 1$$

so

$$(z-2)^3 - 1 = 0$$

Expanding:

$$\begin{aligned}(z-2)^3 - 1 &= (z^3 - 6z^2 + 12z - 8) - 1 \\ &= z^3 - 6z^2 + 12z - 9\end{aligned}$$

So $P, Q, R$ are the roots of

$$z^3 - 6z^2 + 12z - 9 = 0$$

Therefore a degree 6 polynomial with roots $A, B, C, P, Q, R$ is

$$(z^3 - 1)(z^3 - 6z^2 + 12z - 9) = 0$$

Now expand:

$$\begin{aligned}(z^3 - 1)(z^3 - 6z^2 + 12z - 9) &= z^3(z^3 - 6z^2 + 12z - 9) - (z^3 - 6z^2 + 12z - 9) \\ &= z^6 - 6z^5 + 12z^4 - 9z^3 - z^3 + 6z^2 - 12z + 9 \\ &= z^6 - 6z^5 + 12z^4 - 10z^3 + 6z^2 - 12z + 9\end{aligned}$$

Hence the required polynomial equation is

$$z^6 - 6z^5 + 12z^4 - 10z^3 + 6z^2 - 12z + 9 = 0$$

**6.** In an Argand diagram, the points $A$, $B$ and $C$ are the vertices of an equilateral triangle with centre $G$. The point $G$ represents the complex number $1-2i$, and the point $A$ represents the complex number $5+2i$.

**(a)** Find the complex numbers represented by the points $B$ and $C$, giving your answers in the form $x+iy$, where $x$ and $y$ are real and exact. **[6]**

The points $D$, $E$ and $F$ are the centroids of triangles $GBC$, $GCA$ and $GAB$ respectively.

**(b)** Find the exact area of triangle $DEF$. **[4]**

**Solution**

**(a)** Let

$$z_G = 1-2i, \qquad z_A = 5+2i$$

Then the vector from $G$ to $A$ is

$$z_A - z_G = (5+2i)-(1-2i) = 4+4i$$

In an equilateral triangle, the vectors from the centre to the three vertices are $120°$ apart. So the vectors from $G$ to $B$ and $C$ are obtained by rotating $4+4i$ through $\pm 120°$.

Take $B$ to be the $+120°$ rotation. Then

$$z_B - z_G = (4+4i)\left(-\frac{1}{2}+\frac{\sqrt{3}}{2}i\right)$$

Expanding,

$$\begin{aligned} z_B - z_G &= 4\left(-\frac{1}{2}\right)+4\left(\frac{\sqrt{3}}{2}i\right)+4i\left(-\frac{1}{2}\right)+4i\left(\frac{\sqrt{3}}{2}i\right) \\ &= -2+2\sqrt{3}\,i-2i+2\sqrt{3}\,i^2 \\ &= -2-2\sqrt{3}+(2\sqrt{3}-2)i \end{aligned}$$

Hence

$$\begin{aligned} z_B &= z_G+(z_B-z_G) \\ &= (1-2i)+\left[-2-2\sqrt{3}+(2\sqrt{3}-2)i\right] \\ &= -1-2\sqrt{3}+(2\sqrt{3}-4)i \end{aligned}$$

Now take $C$ to be the $-120°$ rotation. Then

$$z_C - z_G = (4+4i)\left(-\frac{1}{2}-\frac{\sqrt{3}}{2}i\right)$$

Expanding,

$$\begin{aligned} z_C - z_G &= 4\left(-\frac{1}{2}\right)+4\left(-\frac{\sqrt{3}}{2}i\right)+4i\left(-\frac{1}{2}\right)+4i\left(-\frac{\sqrt{3}}{2}i\right) \\ &= -2-2\sqrt{3}\,i-2i-2\sqrt{3}\,i^2 \\ &= -2+2\sqrt{3}-(2+2\sqrt{3})i \end{aligned}$$

Hence

$$\begin{aligned} z_C &= z_G+(z_C-z_G) \\ &= (1-2i)+\left[-2+2\sqrt{3}-(2+2\sqrt{3})i\right] \\ &= -1+2\sqrt{3}-(4+2\sqrt{3})i \end{aligned}$$

Therefore

$$B = -1 - 2\sqrt{3} + (2\sqrt{3} - 4)i, \qquad C = -1 + 2\sqrt{3} - (4 + 2\sqrt{3})i$$

with $B$ and $C$ interchangeable.

**(b)** Since $G$ is the centre of the equilateral triangle $ABC$, it is also the centroid of $ABC$. Therefore

$$z_G = \frac{z_A + z_B + z_C}{3}$$

so

$$z_B + z_C = 3z_G - z_A$$

The centroid of triangle $GBC$ is $D$, so

$$\begin{aligned} z_D &= \frac{z_G + z_B + z_C}{3} \\ &= \frac{z_G + (3z_G - z_A)}{3} \\ &= \frac{4z_G - z_A}{3} \end{aligned}$$

Similarly,

$$z_E = \frac{4z_G - z_B}{3}, \qquad z_F = \frac{4z_G - z_C}{3}$$

Now compare these with $z_G$:

$$\begin{aligned} z_D - z_G &= \frac{4z_G - z_A}{3} - z_G = \frac{z_G - z_A}{3} = -\frac{1}{3}(z_A - z_G) \\ z_E - z_G &= -\frac{1}{3}(z_B - z_G) \\ z_F - z_G &= -\frac{1}{3}(z_C - z_G) \end{aligned}$$

So triangle $DEF$ is similar to triangle $ABC$, with all lengths scaled by a factor of $\frac{1}{3}$. Hence the area scale factor is

$$\left(\frac{1}{3}\right)^2 = \frac{1}{9}$$

Next, find the area of $ABC$.

First,

$$GA = |z_A - z_G| = |4 + 4i| = \sqrt{4^2 + 4^2} = 4\sqrt{2}$$

Since $G$ is the centre of the equilateral triangle, the three angles at $G$ are equal and sum to $360°$, so

$$\angle AGB = 120°$$

Also,

$$GA = GB = GC = 4\sqrt{2}$$

Hence

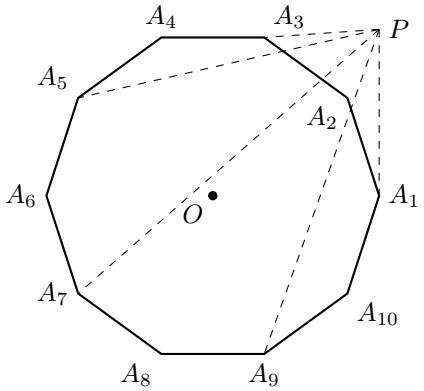
$$\begin{aligned} \text{Area of } \triangle AGB &= \frac{1}{2}(GA)(GB)\sin 120° \\ &= \frac{1}{2}(4\sqrt{2})(4\sqrt{2}) \cdot \frac{\sqrt{3}}{2} \\ &= 16 \cdot \frac{\sqrt{3}}{2} \\ &= 8\sqrt{3} \end{aligned}$$

The triangle $ABC$ is made up of the three congruent triangles $AGB$, $BGC$ and $CGA$, so

$$\text{Area of } ABC = 3(8\sqrt{3}) = 24\sqrt{3}$$

Therefore

$$\begin{aligned} \text{Area of } DEF &= \frac{1}{9} \times 24\sqrt{3} \\ &= \frac{8\sqrt{3}}{3} \end{aligned}$$




**7.** **(a)** Using the identity $zz^* = |z|^2$, or otherwise, show that if $w$ is any root of unity then

$$|w - (1+i)|^2 = 3 - (1-i)w - (1+i)w^* \qquad \textbf{[3]}$$

The diagram shows a regular decagon $A_1A_2 \ldots A_{10}$ whose vertices all lie on the unit circle with centre at the origin $O$ and $A_1$ at $(1, 0)$. The point $P$ lies in the same plane as the decagon and has coordinates $(1, 1)$.

**(b)** Using the result from part (a), find

$$\sum_{r=1}^{5} (PA_{2r-1})^2 \qquad \textbf{[4]}$$

**Solution**

**(a)** Using

$$|z|^2 = zz^*$$

we have

$$\begin{aligned} |w - (1+i)|^2 &= \big(w - (1+i)\big)\big(w^* - (1-i)\big) \\ &= ww^* - (1-i)w - (1+i)w^* + (1+i)(1-i) \end{aligned}$$

Now $w$ is a root of unity, so $|w| = 1$. Hence

$$ww^* = |w|^2 = 1$$

Also,

$$(1+i)(1-i) = 1 - i^2 = 2$$

So

$$\begin{aligned} |w - (1+i)|^2 &= 1 - (1-i)w - (1+i)w^* + 2 \\ &= 3 - (1-i)w - (1+i)w^* \end{aligned}$$

Therefore

$$|w - (1+i)|^2 = 3 - (1-i)w - (1+i)w^*$$

**(b)** Let the complex number representing $A_{2r-1}$ be $w_r$.
Since the decagon is regular and $A_1 = (1, 0)$, the odd-numbered vertices are at arguments

$$0, \ \frac{2\pi}{5}, \ \frac{4\pi}{5}, \ \frac{6\pi}{5}, \ \frac{8\pi}{5}$$

so

$$w_r = e^{2\pi i(r-1)/5} \qquad (r = 1, 2, 3, 4, 5)$$

These are exactly the 5th roots of unity.

Also, the point $P = (1, 1)$ corresponds to the complex number $1 + i$. Therefore

$$PA_{2r-1} = |w_r - (1 + i)|$$

and so

$$(PA_{2r-1})^2 = |w_r - (1 + i)|^2$$

Using the result from part (a),

$$(PA_{2r-1})^2 = 3 - (1 - i)w_r - (1 + i)w_r^*$$

Now sum for $r = 1$ to 5:

$$\begin{aligned}\sum_{r=1}^{5}(PA_{2r-1})^2 &= \sum_{r=1}^{5}(3 - (1 - i)w_r - (1 + i)w_r^*) \\ &= 15 - (1 - i)\sum_{r=1}^{5} w_r - (1 + i)\sum_{r=1}^{5} w_r^*\end{aligned}$$

Since $w_1, w_2, w_3, w_4, w_5$ are the 5th roots of unity,

$$\sum_{r=1}^{5} w_r = 0$$

Taking complex conjugates gives

$$\sum_{r=1}^{5} w_r^* = 0$$

Hence

$$\sum_{r=1}^{5}(PA_{2r-1})^2 = 15$$

**8.** You are given the complex number $\alpha = \cos\frac{3\pi}{5} + i\sin\frac{3\pi}{5}$ and the equation $z^5 = -1$.

**(a)** Show that $\alpha$ is a root of the equation. **[2]**

**(b)** Write down the other four roots of the equation. **[1]**

**(c)** Show that
$$\alpha^4 - \alpha^3 + \alpha^2 - \alpha + 1 = 0$$ **[2]**

**(d)** Hence show that
$$\left(\alpha + \frac{1}{\alpha}\right)^2 - \left(\alpha + \frac{1}{\alpha}\right) - 1 = 0$$ **[3]**

**(e)** Hence determine the value of $\cos\frac{3\pi}{5}$ in the form $a + b\sqrt{c}$, where $a$, $b$ and $c$ are rational numbers to be found. **[4]**

**Solution**

**(a)** We have
$$\alpha = \cos\frac{3\pi}{5} + i\sin\frac{3\pi}{5}$$

Using De Moivre's theorem,
$$\begin{aligned}\alpha^5 &= \cos\left(5\cdot\frac{3\pi}{5}\right) + i\sin\left(5\cdot\frac{3\pi}{5}\right)\\ &= \cos 3\pi + i\sin 3\pi\\ &= -1 + 0i\\ &= -1\end{aligned}$$

So $\alpha$ satisfies $z^5 = -1$, and hence $\alpha$ is a root of the equation.

**(b)** Since
$$-1 = \cos\big((2k+1)\pi\big) + i\sin\big((2k+1)\pi\big)$$
the five roots are
$$z = \cos\frac{(2k+1)\pi}{5} + i\sin\frac{(2k+1)\pi}{5} \qquad (k = 0, 1, 2, 3, 4)$$

Given that $\alpha = \cos\frac{3\pi}{5} + i\sin\frac{3\pi}{5}$, the other four roots are
$$\cos\frac{\pi}{5} + i\sin\frac{\pi}{5}, \quad \cos\pi + i\sin\pi, \quad \cos\frac{7\pi}{5} + i\sin\frac{7\pi}{5}, \quad \cos\frac{9\pi}{5} + i\sin\frac{9\pi}{5}$$

**(c)** From part (a), $\alpha^5 = -1$, so
$$\alpha^5 + 1 = 0$$

Divide $x^5 + 1$ by $x + 1$ to obtain the factorisation
$$x^5 + 1 = (x+1)(x^4 - x^3 + x^2 - x + 1)$$

Hence
$$(\alpha+1)(\alpha^4 - \alpha^3 + \alpha^2 - \alpha + 1) = 0$$

But $\alpha \neq -1$, since $\alpha = \cos\frac{3\pi}{5} + i\sin\frac{3\pi}{5}$ and $\sin\frac{3\pi}{5} \neq 0$.
Therefore
$$\alpha^4 - \alpha^3 + \alpha^2 - \alpha + 1 = 0$$

**(d)** Since $\alpha \neq 0$, divide the result from part (c) by $\alpha^2$:

$$\frac{\alpha^4}{\alpha^2} - \frac{\alpha^3}{\alpha^2} + \frac{\alpha^2}{\alpha^2} - \frac{\alpha}{\alpha^2} + \frac{1}{\alpha^2} = 0$$
$$\alpha^2 - \alpha + 1 - \frac{1}{\alpha} + \frac{1}{\alpha^2} = 0$$

Let

$$t = \alpha + \frac{1}{\alpha}$$

Then

$$t^2 = \left(\alpha + \frac{1}{\alpha}\right)^2$$
$$= \alpha^2 + 2 + \frac{1}{\alpha^2}$$

so

$$\alpha^2 + \frac{1}{\alpha^2} = t^2 - 2$$

Also,

$$\alpha + \frac{1}{\alpha} = t$$

Substituting into

$$\alpha^2 - \alpha + 1 - \frac{1}{\alpha} + \frac{1}{\alpha^2} = 0$$

gives

$$\left(\alpha^2 + \frac{1}{\alpha^2}\right) - \left(\alpha + \frac{1}{\alpha}\right) + 1 = 0$$
$$(t^2 - 2) - t + 1 = 0$$
$$t^2 - t - 1 = 0$$

Therefore,

$$\left(\alpha + \frac{1}{\alpha}\right)^2 - \left(\alpha + \frac{1}{\alpha}\right) - 1 = 0$$

**(e)** From part (d), let

$$t = \alpha + \frac{1}{\alpha}$$

so that

$$t^2 - t - 1 = 0$$

Solving this quadratic,

$$t = \frac{1 \pm \sqrt{5}}{2}$$

Now

$$\alpha = \cos\frac{3\pi}{5} + i\sin\frac{3\pi}{5}$$

and since $|\alpha| = 1$,

$$\frac{1}{\alpha} = \cos\frac{3\pi}{5} - i\sin\frac{3\pi}{5}$$

Hence

$$\alpha + \frac{1}{\alpha} = \left(\cos\frac{3\pi}{5} + i\sin\frac{3\pi}{5}\right) + \left(\cos\frac{3\pi}{5} - i\sin\frac{3\pi}{5}\right)$$
$$= 2\cos\frac{3\pi}{5}$$

So

$$2\cos\frac{3\pi}{5} = \frac{1 \pm \sqrt{5}}{2}$$

Therefore

$$\cos\frac{3\pi}{5} = \frac{1 \pm \sqrt{5}}{4}$$

Since $\frac{3\pi}{5} = 108°$ is in quadrant II, $\cos\frac{3\pi}{5} < 0$, so we take the negative value:

$$\cos\frac{3\pi}{5} = \frac{1 - \sqrt{5}}{4} = \frac{1}{4} - \frac{1}{4}\sqrt{5}$$

So $\cos\frac{3\pi}{5}$ is in the form $a + b\sqrt{c}$ with $a = \frac{1}{4}$, $b = -\frac{1}{4}$ and $c = 5$.

**9.** The points representing the complex numbers $z_1 = 6 + 11i$ and $z_2 = -8 - 3i$ are opposite vertices of a regular octagon, $K$, in the complex plane.
The centre of $K$ represents the complex number $\alpha$.

**(a)** Show that $\alpha = -1 + 4i$. **[2]**

Given that

$$\beta = \frac{1 - i}{14}$$

**(b)** Show that

$$\beta(z_1 - \alpha) = 1$$ **[2]**

The vertices of $K$ are given by the roots of the equation

$$(\beta(z - \alpha))^8 = 1$$

**(c)** (i) Write down the roots of the equation $w^8 = 1$ in the form $re^{i\theta}$. **[1]**

(ii) Hence, or otherwise, determine the positions of the other six vertices of $K$, giving your answers as complex numbers in Cartesian form. **[4]**

**Solution**

**(a)** Since $z_1$ and $z_2$ are opposite vertices of the regular octagon, the centre is the midpoint of the line joining them.

$$\alpha = \frac{z_1 + z_2}{2} = \frac{(6 + 11i) + (-8 - 3i)}{2}$$

$$\alpha = \frac{-2 + 8i}{2} = -1 + 4i$$

So $\alpha = -1 + 4i$.

**(b)** First find $z_1 - \alpha$:

$$z_1 - \alpha = (6 + 11i) - (-1 + 4i) = 7 + 7i = 7(1 + i)$$

Now multiply by $\beta$:

$$\beta(z_1 - \alpha) = \frac{1 - i}{14} \cdot 7(1 + i) = \frac{(1 - i)(1 + i)}{2}$$

Using $(1 - i)(1 + i) = 1 - i^2 = 2$,

$$\beta(z_1 - \alpha) = \frac{2}{2} = 1$$

Hence $\beta(z_1 - \alpha) = 1$.

**(c)** (i) The roots of $w^8 = 1$ are the eighth roots of unity:

$$w = e^{ik\pi/4}, \qquad k = 0, 1, 2, 3, 4, 5, 6, 7$$

So in the form $re^{i\theta}$, we have $r = 1$ and $\theta = \frac{k\pi}{4}$.

(ii) From

$$(\beta(z-\alpha))^8 = 1$$

let

$$w = \beta(z-\alpha)$$

Then $w^8 = 1$, so

$$z - \alpha = \frac{w}{\beta} \qquad \Rightarrow \qquad z = \alpha + \frac{w}{\beta}$$

Now

$$\frac{1}{\beta} = \frac{14}{1-i} = \frac{14(1+i)}{(1-i)(1+i)} = \frac{14(1+i)}{2} = 7(1+i)$$

Hence

$$z = \alpha + 7(1+i)w$$

Using $\alpha = -1 + 4i$,

$$z = -1 + 4i + 7(1+i)w$$

Now substitute the eight roots of $w^8 = 1$.
For $w = 1$,

$$z = -1 + 4i + 7(1+i) = 6 + 11i$$

which is $z_1$.
For $w = -1$,

$$z = -1 + 4i - 7(1+i) = -8 - 3i$$

which is $z_2$.
So the other six vertices come from the remaining six values of $w$.
For $w = e^{i\pi/4} = \dfrac{1+i}{\sqrt{2}}$,

$$z = -1 + 4i + 7(1+i)\frac{1+i}{\sqrt{2}} = -1 + 4i + 7\sqrt{2}\,i$$

so

$$z = -1 + (4 + 7\sqrt{2})i$$

For $w = e^{i\pi/2} = i$,

$$z = -1 + 4i + 7(1+i)i = -1 + 4i + 7(-1+i)$$

so

$$z = -8 + 11i$$

For $w = e^{3i\pi/4} = \dfrac{-1+i}{\sqrt{2}}$,

$$z = -1 + 4i + 7(1+i)\frac{-1+i}{\sqrt{2}} = -1 + 4i - 7\sqrt{2}$$

so

$$z = (-1 - 7\sqrt{2}) + 4i$$

For $w = e^{5i\pi/4} = \dfrac{-1-i}{\sqrt{2}}$,

$$z = -1 + 4i + 7(1+i)\frac{-1-i}{\sqrt{2}} = -1 + 4i - 7\sqrt{2}\,i$$

so

$$z = -1 + (4 - 7\sqrt{2})i$$

For $w = e^{3i\pi/2} = -i$,

$$z = -1 + 4i + 7(1+i)(-i) = -1 + 4i + 7(1-i)$$

so

$$z = 6 - 3i$$

For $w = e^{7i\pi/4} = \dfrac{1-i}{\sqrt{2}}$,

$$z = -1 + 4i + 7(1+i)\frac{1-i}{\sqrt{2}} = -1 + 4i + 7\sqrt{2}$$

so

$$z = (-1 + 7\sqrt{2}) + 4i$$

Therefore the other six vertices are

$$-1 + (4 + 7\sqrt{2})i, \quad -8 + 11i, \quad (-1 - 7\sqrt{2}) + 4i$$
$$-1 + (4 - 7\sqrt{2})i, \quad 6 - 3i, \quad (-1 + 7\sqrt{2}) + 4i$$

**10.** (i) Find the 8th roots of $-128 + 128\sqrt{3}\,i$ in the form $re^{i\theta}$, where $r > 0$ and $0 \leq \theta < 2\pi$. **[5]**

(ii) These 8th roots are represented in the Argand diagram, in order of increasing $\theta$, by the points $A$, $B$, $C$, $D$, $E$, $F$, $G$, $H$.
Draw the Argand diagram, making clear which point is which. **[2]**

(iii) The mid-point of $BD$ is the point $P$ which represents the complex number $w$.
Find, in exact form, the modulus and argument of $w$. **[3]**

(iv) $w$ is an $n$th root of a real number $a$, where $n$ is a positive integer. State the least possible value of $n$ and find the corresponding value of $a$. **[3]**

**Solution**

**(i)** Let

$$z = -128 + 128\sqrt{3}\,i$$

Its modulus is

$$\begin{aligned} |z| &= \sqrt{(-128)^2 + (128\sqrt{3})^2} \\ &= 128\sqrt{1+3} \\ &= 256 \end{aligned}$$

Also,

$$\tan\theta = \frac{128\sqrt{3}}{-128} = -\sqrt{3}$$

Since $z$ lies in the second quadrant, its argument is

$$\theta = \frac{2\pi}{3}$$

So

$$-128 + 128\sqrt{3}\,i = 256e^{i2\pi/3}$$

If $z_k$ is an 8th root, then

$$z_k^8 = 256e^{i2\pi/3}$$

Hence the modulus of each root is

$$256^{1/8} = 2$$

and the arguments are

$$\frac{\frac{2\pi}{3} + 2k\pi}{8} = \frac{\pi}{12} + \frac{k\pi}{4}, \qquad k = 0, 1, \ldots, 7$$

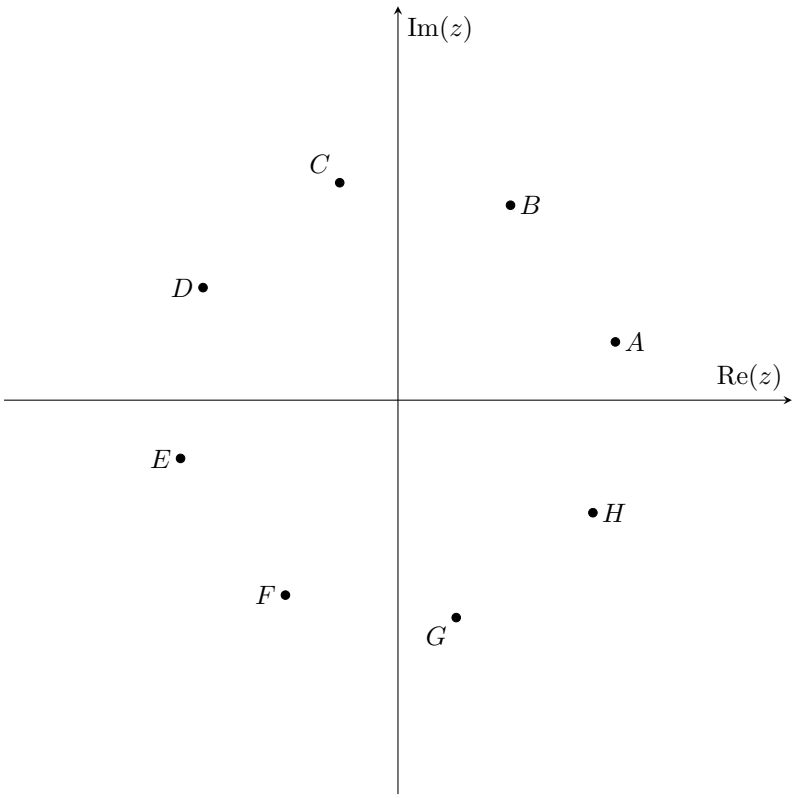
Therefore the 8th roots are

$$z_k = 2e^{i(\pi/12 + k\pi/4)}, \qquad k = 0, 1, \ldots, 7$$

**(ii)** In increasing order of argument, the points are

$$A : \frac{\pi}{12}, \quad B : \frac{\pi}{3}, \quad C : \frac{7\pi}{12}, \quad D : \frac{5\pi}{6}, \quad E : \frac{13\pi}{12}, \quad F : \frac{4\pi}{3}, \quad G : \frac{19\pi}{12}, \quad H : \frac{11\pi}{6}$$

All eight points lie on the circle $|z| = 2$.




**(iii)** The complex numbers represented by $B$ and $D$ are

$$2e^{i\pi/3} \quad \text{and} \quad 2e^{i5\pi/6}$$

Since $P$ is the midpoint of $BD$, the complex number $w$ is

$$\begin{aligned} w &= \frac{2e^{i\pi/3} + 2e^{i5\pi/6}}{2} \\ &= e^{i\pi/3} + e^{i5\pi/6} \\ &= e^{i7\pi/12}\left(e^{-i\pi/4} + e^{i\pi/4}\right) \\ &= e^{i7\pi/12}\left(2\cos\frac{\pi}{4}\right) \\ &= \sqrt{2}\, e^{i7\pi/12} \end{aligned}$$

So the modulus and argument of $w$ are

$$|w| = \sqrt{2}, \qquad \arg(w) = \frac{7\pi}{12}$$

**(iv)** Since

$$w = \sqrt{2}\, e^{i7\pi/12}$$

we have

$$w^n = (\sqrt{2})^n e^{i7n\pi/12}$$

For $w^n$ to be real, its argument must be a multiple of $\pi$, so

$$\frac{7n\pi}{12} = m\pi$$

for some integer $m$. Hence

$$\frac{7n}{12} \in \mathbb{Z}$$

As 7 and 12 have no common factor, the least possible value is

$$n = 12$$

Then

$$\begin{aligned} a &= w^{12} \\ &= (\sqrt{2})^{12} e^{i7\pi} \\ &= 2^6(-1) \\ &= -64 \end{aligned}$$

So the least possible value of $n$ is 12, and the corresponding value of $a$ is $-64$.

**11.** The cubic equation

$$z^3 + pz^2 + qz + 24 = 0$$

where $p$ and $q$ are real, has roots $\alpha$, $\beta$ and $\gamma$.
It is given that $\alpha = -2 + 2i$.

**(a)** (i) Write down another root, $\beta$, of the equation. **[1]**

(ii) Find the third root, $\gamma$. **[3]**

(iii) Find the values of $p$ and $q$. **[3]**

**(b)** (i) Express $\alpha$ in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$. **[2]**

(ii) Hence show that

$$(-2 + 2i)^n = (2\sqrt{2})^n \left( \cos \frac{3n\pi}{4} + i \sin \frac{3n\pi}{4} \right)$$ **[2]**

(iii) Hence show that

$$\alpha^n + \beta^n + \gamma^n = 2(2\sqrt{2})^n \cos \frac{3n\pi}{4} + (-3)^n$$

where $n$ is an integer. **[3]**

**Solution**

**(a)** (i) Since the coefficients of the cubic are real, complex roots occur in conjugate pairs.
So the other complex root is

$$\beta = -2 - 2i$$

(ii) For the monic cubic

$$z^3 + pz^2 + qz + 24 = 0$$

the product of the roots is

$$\alpha\beta\gamma = -24$$

First find $\alpha\beta$:

$$\begin{aligned} \alpha\beta &= (-2 + 2i)(-2 - 2i) \\ &= (-2)^2 - (2i)^2 \\ &= 4 - (-4) \\ &= 8 \end{aligned}$$

So

$$8\gamma = -24$$

and hence

$$\gamma = -3$$

(iii) Using the relationships between roots and coefficients:

$$\alpha + \beta + \gamma = -p$$

Now

$$\begin{aligned} \alpha + \beta + \gamma &= (-2 + 2i) + (-2 - 2i) + (-3) \\ &= -7 \end{aligned}$$

so

$$-p = -7$$

giving

$$p = 7$$

Also

$$\alpha\beta + \beta\gamma + \gamma\alpha = q$$

We already know that $\alpha\beta = 8$, and

$$\begin{aligned}\beta\gamma + \gamma\alpha &= \gamma(\alpha + \beta)\\ &= (-3)\big((-2+2i) + (-2-2i)\big)\\ &= (-3)(-4)\\ &= 12\end{aligned}$$

Therefore

$$q = 8 + 12 = 20$$

So the values are

$$p = 7, \qquad q = 20$$

**(b)** (i) For

$$\alpha = -2 + 2i$$

its modulus is

$$\begin{aligned}r &= |\alpha|\\ &= \sqrt{(-2)^2 + 2^2}\\ &= \sqrt{8}\\ &= 2\sqrt{2}\end{aligned}$$

The point $(-2, 2)$ lies in the second quadrant, so the principal argument is

$$\theta = \frac{3\pi}{4}$$

Hence

$$\alpha = 2\sqrt{2}\, e^{i3\pi/4}$$

(ii) From part (i),

$$\alpha = 2\sqrt{2}\, e^{i3\pi/4}$$

so

$$\begin{aligned}\alpha^n &= \left(2\sqrt{2}\, e^{i3\pi/4}\right)^n\\ &= (2\sqrt{2})^n e^{i3n\pi/4}\end{aligned}$$

Using $e^{i\theta} = \cos\theta + i\sin\theta$,

$$(-2+2i)^n = (2\sqrt{2})^n \left(\cos\frac{3n\pi}{4} + i\sin\frac{3n\pi}{4}\right)$$

(iii) Since

$$\beta = -2 - 2i = 2\sqrt{2}\, e^{-i3\pi/4}$$

we have

$$\begin{aligned}\beta^n &= \left(2\sqrt{2}\, e^{-i3\pi/4}\right)^n\\ &= (2\sqrt{2})^n \left(\cos\left(-\frac{3n\pi}{4}\right) + i\sin\left(-\frac{3n\pi}{4}\right)\right)\\ &= (2\sqrt{2})^n \left(\cos\frac{3n\pi}{4} - i\sin\frac{3n\pi}{4}\right)\end{aligned}$$

Therefore

$$\begin{aligned}\alpha^n + \beta^n &= (2\sqrt{2})^n \left(\cos\frac{3n\pi}{4} + i\sin\frac{3n\pi}{4}\right) + (2\sqrt{2})^n \left(\cos\frac{3n\pi}{4} - i\sin\frac{3n\pi}{4}\right) \\ &= 2(2\sqrt{2})^n \cos\frac{3n\pi}{4}\end{aligned}$$

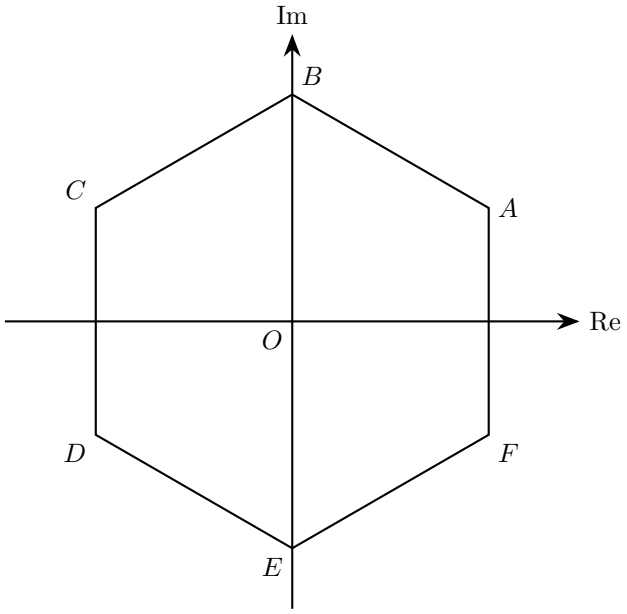
Also, from part (a), $\gamma = -3$, so

$$\gamma^n = (-3)^n$$

Hence

$$\alpha^n + \beta^n + \gamma^n = 2(2\sqrt{2})^n \cos\frac{3n\pi}{4} + (-3)^n$$

as required.




**12.** In the diagram, the points $A$, $B$, $C$, $D$, $E$ and $F$ represent the complex sixth roots of $-729$ on an Argand diagram. The centroids of triangles $OAB$, $OBC$, $OCD$, $ODE$, $OEF$ and $OFA$ are $G$, $H$, $I$, $J$, $K$ and $L$ respectively.

**(a)** Write down, in exponential form, the complex numbers represented by the points $A$, $B$, $C$, $D$, $E$ and $F$. **[2]**

**(b)** When these complex numbers are multiplied by the complex number $w$, the resulting complex numbers are represented by the points $G$, $H$, $I$, $J$, $K$ and $L$.
Find $w$ in exponential form. **[4]**

**(c)** You are given that $G$, $H$, $I$, $J$, $K$ and $L$ represent roots of the equation $z^6 = p$.
Find $p$. **[2]**

**Solution**

**(a)** Since

$$-729 = 729e^{i\pi} = 3^6 e^{i(\pi+2k\pi)}$$

the sixth roots are

$$z = 3e^{i\left(\frac{\pi+2k\pi}{6}\right)} = 3e^{i\left(\frac{\pi}{6}+\frac{k\pi}{3}\right)}, \qquad k = 0, 1, 2, 3, 4, 5$$

Taking these in order round the Argand diagram gives

$$A = 3e^{i\pi/6}, \qquad B = 3e^{i\pi/2}, \qquad C = 3e^{i5\pi/6}$$
$$D = 3e^{i7\pi/6}, \qquad E = 3e^{i3\pi/2}, \qquad F = 3e^{i11\pi/6}$$

**(b)** The centroid of triangle $OAB$ is

$$G = \frac{O + A + B}{3} = \frac{A + B}{3}$$

If multiplication by $w$ maps $A$ to $G$, then

$$Aw = G = \frac{A + B}{3}$$

so

$$w = \frac{A + B}{3A}$$

Now $B$ is the next sixth root after $A$, so

$$B = Ae^{i\pi/3}$$

Hence

$$\begin{aligned} w &= \frac{A + Ae^{i\pi/3}}{3A} \\ &= \frac{1 + e^{i\pi/3}}{3} \end{aligned}$$

Now

$$e^{i\pi/3} = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

so

$$1 + e^{i\pi/3} = \frac{3}{2} + \frac{\sqrt{3}}{2}i$$

Its modulus is

$$\sqrt{\left(\frac{3}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} = \sqrt{\frac{9}{4} + \frac{3}{4}} = \sqrt{3}$$

and its argument is

$$\tan^{-1}\left(\frac{\sqrt{3}/2}{3/2}\right) = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$$

So

$$1 + e^{i\pi/3} = \sqrt{3}\, e^{i\pi/6}$$

Therefore

$$w = \frac{\sqrt{3}}{3}e^{i\pi/6} = \frac{1}{\sqrt{3}}e^{i\pi/6}$$

Also, for example,

$$H = \frac{B + C}{3} = \frac{B + Be^{i\pi/3}}{3} = Bw$$

and similarly for the other consecutive pairs, so this same $w$ maps $A, B, C, D, E, F$ to $G, H, I, J, K, L$.

Hence

$$w = \frac{1}{\sqrt{3}}e^{i\pi/6}$$

**(c)** Let $r$ be any sixth root of $-729$. Then the corresponding point among $G, H, I, J, K, L$ is

$$z = wr$$

So

$$z^6 = w^6 r^6 = w^6(-729)$$

Now

$$\begin{aligned} w^6 &= \left(\frac{1}{\sqrt{3}}e^{i\pi/6}\right)^6 \\ &= \left(\frac{1}{\sqrt{3}}\right)^6 e^{i\pi} \\ &= \frac{1}{27}(-1) \\ &= -\frac{1}{27} \end{aligned}$$

Hence

$$p = w^6(-729) = \left(-\frac{1}{27}\right)(-729) = 27$$

**13.** (i) Write down, in any form, all the complex roots of the equation

$$w^{11} = -1$$ **[2]**

(ii) Explain why the equation

$$(z+1)^{11} = -(z-1)^{11}$$

has exactly 10 non-real roots and show that they may be expressed in the form

$$-i\cot\left(\frac{m\pi}{22}\right)$$ **[6]**

where $m = \pm 1, \pm 3, \pm 5, \pm 7, \pm 9$.

(iii) Show that

$$\left\{-i\cot\left(\frac{m\pi}{22}\right)\right\}\left\{i\cot\left(\frac{m\pi}{22}\right)\right\} = \cot^2\left(\frac{m\pi}{22}\right)$$ **[1]**

(iv) Given that the product of the non-zero roots of the equation in part (ii) is 11, find the value of

$$\tan^2\left(\frac{\pi}{22}\right)\tan^2\left(\frac{3\pi}{22}\right)\tan^2\left(\frac{5\pi}{22}\right)\tan^2\left(\frac{7\pi}{22}\right)\tan^2\left(\frac{9\pi}{22}\right)$$ **[2]**

**Solution**

**(i)** Write $-1$ in exponential form as

$$-1 = e^{i(2k+1)\pi} \qquad (k \in \mathbb{Z})$$

So if $w^{11} = -1$, then

$$w = e^{i(2k+1)\pi/11}$$

Taking the 11 distinct values gives

$$w = e^{i(2k+1)\pi/11}, \qquad k = 0, 1, \ldots, 10$$

So all the complex roots are

$$e^{i\pi/11},\ e^{3i\pi/11},\ e^{5i\pi/11},\ \ldots,\ e^{21i\pi/11}$$

**(ii)** First note that $z = 1$ is not a root of

$$(z+1)^{11} = -(z-1)^{11}$$

because this would give $2^{11} = 0$, which is false.

So we may divide by $(z-1)^{11}$:

$$\left(\frac{z+1}{z-1}\right)^{11} = -1$$

Let

$$w = \frac{z+1}{z-1}$$

Then

$$w^{11} = -1$$

From part (i), the 11 possible values of $w$ are

$$w = e^{i\theta}, \qquad \theta = \frac{(2k+1)\pi}{11}, \qquad k = 0, 1, \ldots, 10$$

Now solve for $z$ in terms of $w$:

$$\begin{aligned} w &= \frac{z+1}{z-1} \\ w(z-1) &= z+1 \\ wz - w &= z+1 \\ z(w-1) &= w+1 \\ z &= \frac{w+1}{w-1} \end{aligned}$$

So

$$z = \frac{e^{i\theta}+1}{e^{i\theta}-1}$$

Now multiply top and bottom by $e^{-i\theta/2}$:

$$\begin{aligned} z &= \frac{e^{i\theta/2}+e^{-i\theta/2}}{e^{i\theta/2}-e^{-i\theta/2}} \\ &= \frac{2\cos(\theta/2)}{2i\sin(\theta/2)} \\ &= -i\cot\left(\frac{\theta}{2}\right) \end{aligned}$$

Since $\theta = \dfrac{(2k+1)\pi}{11}$, this gives

$$z = -i\cot\left(\frac{(2k+1)\pi}{22}\right) \qquad k = 0, 1, \ldots, 10$$

So the 11 roots are obtained from

$$m = 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21$$

in

$$z = -i\cot\left(\frac{m\pi}{22}\right)$$

When $m = 11$,

$$z = -i\cot\left(\frac{11\pi}{22}\right) = -i\cot\left(\frac{\pi}{2}\right) = 0$$

so this is the one real root.

To see there are no other real roots, suppose $z$ is real. Then

$$w = \frac{z+1}{z-1}$$

is also real. But the only real solution of $w^{11} = -1$ is $w = -1$, which gives

$$z = \frac{-1+1}{-1-1} = 0$$

Hence there is exactly one real root, namely 0, so the remaining $11 - 1 = 10$ roots are non-real.

Now rewrite the values with $m = 13, 15, 17, 19, 21$. If $r = 1, 3, 5, 7, 9$, then

$$\begin{aligned} -i\cot\left(\frac{(22-r)\pi}{22}\right) &= -i\cot\left(\pi - \frac{r\pi}{22}\right) \\ &= -i\left(-\cot\left(\frac{r\pi}{22}\right)\right) \\ &= i\cot\left(\frac{r\pi}{22}\right) \\ &= -i\cot\left(-\frac{r\pi}{22}\right) \end{aligned}$$

Therefore the 10 non-real roots may be written as

$$-i\cot\left(\frac{m\pi}{22}\right), \qquad m = \pm 1, \pm 3, \pm 5, \pm 7, \pm 9$$

**(iii)** Let

$$\alpha = \frac{m\pi}{22}$$

Then

$$\begin{aligned}(-i\cot\alpha)(i\cot\alpha) &= (-i\cdot i)\cot^2\alpha \\ &= -i^2\cot^2\alpha \\ &= \cot^2\alpha\end{aligned}$$

So

$$\left\{-i\cot\left(\frac{m\pi}{22}\right)\right\}\left\{i\cot\left(\frac{m\pi}{22}\right)\right\} = \cot^2\left(\frac{m\pi}{22}\right)$$

**(iv)** The non-zero roots occur in pairs

$$-i\cot\left(\frac{m\pi}{22}\right) \quad \text{and} \quad -i\cot\left(-\frac{m\pi}{22}\right) = i\cot\left(\frac{m\pi}{22}\right)$$

for

$$m = 1, 3, 5, 7, 9$$

Given that the product of the non-zero roots is 11,

$$\begin{aligned}11 &= \prod_{m=1,3,5,7,9}\left\{-i\cot\left(\frac{m\pi}{22}\right)\right\}\left\{i\cot\left(\frac{m\pi}{22}\right)\right\} \\ &= \prod_{m=1,3,5,7,9}\cot^2\left(\frac{m\pi}{22}\right)\end{aligned}$$

Therefore

$$\cot^2\left(\frac{\pi}{22}\right)\cot^2\left(\frac{3\pi}{22}\right)\cot^2\left(\frac{5\pi}{22}\right)\cot^2\left(\frac{7\pi}{22}\right)\cot^2\left(\frac{9\pi}{22}\right) = 11$$

Taking reciprocals,

$$\tan^2\left(\frac{\pi}{22}\right)\tan^2\left(\frac{3\pi}{22}\right)\tan^2\left(\frac{5\pi}{22}\right)\tan^2\left(\frac{7\pi}{22}\right)\tan^2\left(\frac{9\pi}{22}\right) = \frac{1}{11}$$